

Basic of Computer Networks 5

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'I think, in general, it's clear that most bad things come from misunderstanding, and communication is generally the way to resolve misunderstandings — and the Web's a form of communications — so it generally should be good.'

Tim-Bernes Lee - Creator of the WEB

- 1 Local Area Networks
 - Introduction
 - Advantages of LANs
 - Disadvantages of LANs
- 2 LAN Topologies
 - Bus Topology
 - Star Topology
 - Ring Topology
- 3 Bus Network
 - The CSMA/CD Paradigm
- 4 Ring Network
 - Token Passing Paradigm
- 5 Wireless LAN
 - The CSMA/CA Paradigm
- 6 Summary

- 1 Local Area Networks
 - Introduction
 - Advantages of LANs
 - Disadvantages of LANs
- 2 LAN Topologies
 - Bus Topology
 - Star Topology
 - Ring Topology
- 3 Bus Network
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- 4 Ring Network
 - Token Passing Paradigm
- 5 Wireless LAN
 - The CSMA/CA Paradigm
- 6 Summary

What is a Local Area Network (or LAN)?

- A local area network is a communication network that interconnects a variety of data communicating devices within a small geographic area and broadcasts data at high data transfer rates with very low error rates
- Since the local area network first appeared in the 70s, its use has become widespread in commercial and academic environments.
- Actually nowadays everybody has a LAN at home (the internet Box)



Advantages of LANs:

- Ability to share hardware and software resources.
- Local computers can survive internet failure and still exchange information each other.
- Computers of the LAN can exchange information at high speed and independently (no need MAN/WAN accesses).
- Private ownership: only the computers belonging to the LAN can access the shared information (if not connected to Internet or properly secured from other networks)
- The error rate is generally much less important.

Disadvantages of LANs:

- You can only access geographically close computers...
- Designing a LAN can sometimes be difficult and costly (especially in industrial environments).
- Some types of hardware may not inter-operate easily.
- A LAN is only as fast as its slowest link!

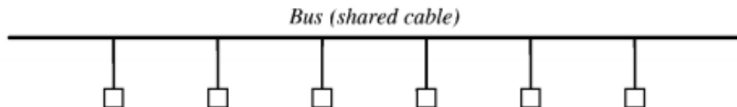
- 1 Local Area Networks
 - Introduction
 - Advantages of LANs
 - Disadvantages of LANs
- 2 LAN Topologies
 - Bus Topology
 - Star Topology
 - Ring Topology
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- 4 Ring Network
 - Token Passing Paradigm
- 5 Wireless LAN
 - The CSMA/CA Paradigm
- 6 Summary

Most Local Area Networks are based on one of these three main topologies:

- ① The Bus Topology
- ② The Star Topology
- ③ The Ring Topology

The Bus Topology:

- Shared medium main interconnect
- Each computer has a connection to the medium

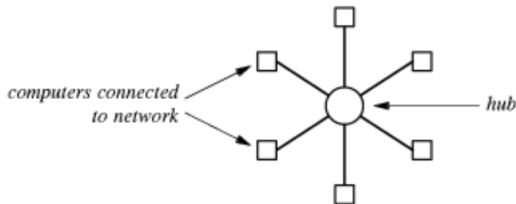


Note

- Easy to add a node to the network (without disturbing the network)
- Resilient to computer failure(s)
- Data transmission management is complex

The Star Topology:

- The central component of the network is known as hub or switch
- Each computer has a separate connection to the hub

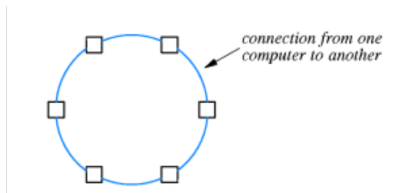


Notes

- If a node (=computer) of the network is disconnected the network still works
- Node to node exchanges are optimized (information simply transit through the switch)

The Ring Topology:

- No central facility (no switch needed)
- Connections go directly from one computer to another
- Invented by IBM in the 80s



Notes

- If one node (=computer) does not work: the ring network is cut!
- Data transmission management can be simplified with this topology

- 1 Local Area Networks
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 - Bus Topology
 - Star Topology
 - Ring Topology
- 3 Bus Network
 - The CSMA/CD Paradigm
- 4 Ring Network
 - Token Passing Paradigm
- 5 Wireless LAN
 - The CSMA/CA Paradigm
- 6 Summary

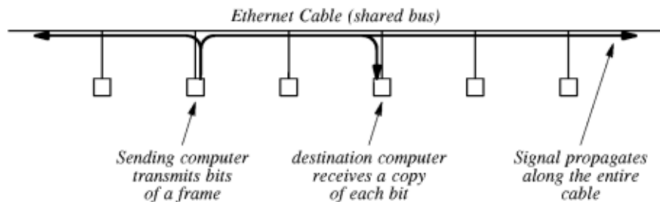
Before talking about Bus Network, let's talk about Ethernet:

Ethernet ...

- ... is a computer network technology
- ... is the most popular wired LAN technology
- ... can have different wiring schemes
- ... can have different data rates (nowadays: up to the gigabyte/sec)
- ... is the fruit of the IEEE 802.3 standard



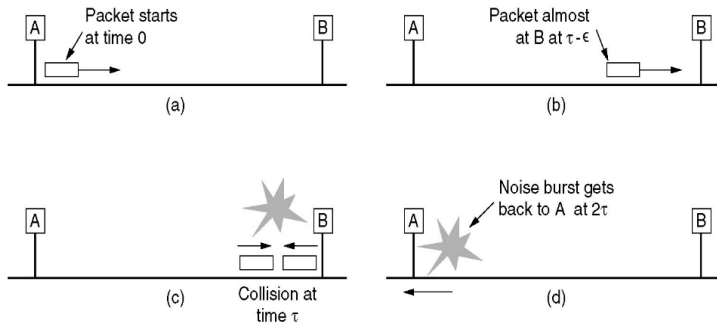
You all know this famous RJ45 Ethernet cable ...



In this case:

- Nodes transmit data using a same shared medium.
- Signal propagates across the entire cable.
- All nodes receive the data.
- Collisions of information may occur! The solution is CSMA/CD.

Some collisions may occur with the Bus Topology, example:



Notes

- Collision detection can take as long as 2 times one-way propagation delay in worst case!
- The CSMA/CD Paradigm is an approach to handle these cases.

But what is the CSMA/CD Paradigm?

- MA: Multiple Access
 - ▶ Multiple computers are attached to a shared medium
 - ▶ Each uses a same access algorithm
- CS: Carrier Sense
 - ▶ Wait until the medium is free
 - ▶ Then transmit the frame
- CD: Collision Detection
- With CSMA/CD: simultaneous transmissions are possible!

How does it work?

When two transmissions interfere with one another we have a collision.

With the CSMA/CD Paradigm the node ...

- ... listens to the medium during transmission
- ... detects whether another node's signal interferes
 - ▶ If the emitting node receives another signal during the transmission of its signal, it means there is a collision
- ... transmits again the signal in case of collision detection (back off)

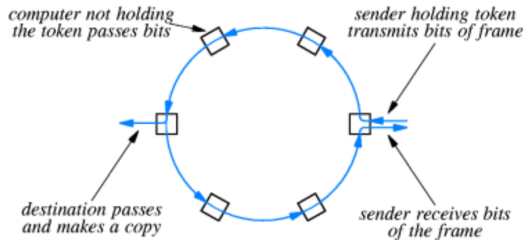
When do we decide to re-transmit the signal with the CSMA/CD Paradigm?

- When a collision occurs: it waits a random time t_1 ($0 \leq t_1 \leq d$)
- If a second collision occurs afterwards: it waits a random time t_2 ($0 \leq t_2 \leq 2 \times d$)
- For each successive collision: the range is doubled.
- It is called: exponential backoff

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- 6 Summary

The Ring Network:

- Once a popular topology for LANs
- Bits flow in single direction



Note

A token is transmitted from one node to node. The node having the token can transmit the data.

The token passing paradigm:

- It is designed for ring topology
- It guarantees fair access
- Token:
 - ▶ It is a special message (reserved)
 - ▶ It's just small (a few bits)

How it works:

- Each node:
 - ▶ waits for token to arrive
 - ▶ transmits one packet around ring
 - ▶ transmits token around ring
- When no node has data to send
 - ▶ token circulates continuously

Token ring topology pros and cons:

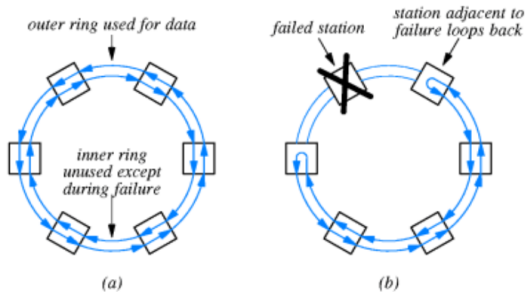
Pros:

- Intermediate hardware (switch or hub) not needed
- A better management of packet transmission (no need of CSMA/CD or CSMA/CA)

Cons

- A broken wire disables the ring!
- Difficult to add/move/remove node with this topology.

Some token ring approaches use two rings (one inner ring and one outer ring) to be resilient in case one node is malfunctioning:



- When working normally one ring is used
- Second ring is used for loopback during failure

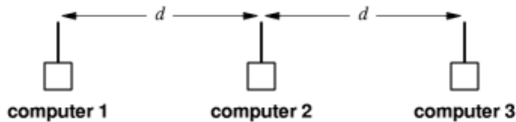
Note

Only one node can fail with this approach!

- 1 Local Area Networks
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 - Disadvantages of LANs
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 - Star Topology
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- 6 Summary

With wireless LANs ...

- ... not all nodes receive all transmissions
- ... cannot use CSMA/CD in this case! Not useful.

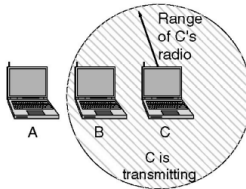


Notes

- Maximum transmission distance is d .
- Nodes 1 and 3 do not receive each other's transmissions.

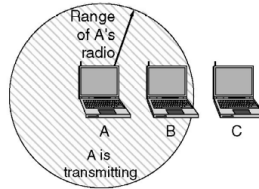
Hidden and Exposed Terminal Effects:

A wants to send to B
but cannot hear that
B is busy



(a)

B wants to send to C
but mistakenly thinks
the transmission will fail



(b)

(a) The hidden station problem. (b) The exposed station problem.

Note

A solution to this is the CSMA/CA Paradigm!

The CSMA/CA Paradigm:

The CSMA/CA Paradigm is based on collision avoidance (and not detection!).

- Used on wireless LANs!
- Both sides send small messages before the data transmission itself!
 - ▶ "X is about to send to Y": Give me some place!!
 - ▶ "Y is about to receive from X": Don't send me things!!
 - ▶ Data frame is finally sent from X to Y
- The purpose is to inform all nodes in range of X or Y before transmission.
- CA: Collision Avoidance

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- 4 Ring Network
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- 6 Summary

- LANs are designed for short distance data exchanges
- Three main topologies: Bus, Ring and Star.
- The most popular LAN technology is Ethernet
- Apart from that, wireless LANs are also very popular
- Although interesting the Token Ring approach is less popular